

Repeated Questions (Set I – Set IV)

1. Knowledge Discovery in Database (KDD)

- Explain phases of KDD (with/without diagram).

2. Data Preprocessing

- Cleaning, integration, transformation, discretization.

3. Correlation & Significance Testing

- Pearson correlation coefficient calculation and interpretation.
- Hypothesis testing (t-test).

4. Regression

- Ordinary Least Squares (OLS) derivation.
- Regression numericals.
- R^2 vs Adjusted R^2 interpretation.

5. Multicollinearity

- Variance Inflation Factor (VIF).

6. Kernel Regression

- Nadaraya-Watson Estimator.

7. Principal Component Analysis (PCA)

- Derivation and dimensionality reduction.

8. Decision Trees / CART Algorithm

- Gini impurity calculations.
- CART methodology.

9. Logistic Regression

- Concept and applications.

10. K-Means Clustering

- Step-by-step iterations.

11. Time Series Analysis

- ARMA/ARIMA models.

- Stationarity, autocorrelation.
- AIC/BIC criteria.

12. Causation vs Correlation

- Granger causality.
- Directed Acyclic Graphs (DAG).

13. Short Notes (Frequently Repeated)

- Empirical Distribution.
 - Statistical Significance (p-value, t-value).
 - Multicollinearity & VIF.
 - Logistic Regression.
 - PCA.
 - Power-law distribution.
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Great — let's highlight the **most high-frequency questions** that appeared in nearly every set (I–IV + model papers). These are the ones you should prioritize for exam prep since they're most likely to be asked again.

High-Frequency Questions (Top Priority)

1. Knowledge Discovery in Database (KDD)

- Define KDD and explain its phases (often with diagram).

2. Data Preprocessing

- Explain cleaning, integration, transformation, and discretization.

3. Correlation & Hypothesis Testing

- Compute Pearson correlation coefficient and interpret results.
- Perform hypothesis testing (t-test).

4. Regression Analysis

- Derive Ordinary Least Squares (OLS) estimator.
- Interpret R^2 vs Adjusted R^2 .
- Regression numericals.

5. Multicollinearity

- Variance Inflation Factor (VIF) and its implications.

6. Kernel Regression

- Nadaraya-Watson estimator.

7. Principal Component Analysis (PCA)

- Derivation and dimensionality reduction.
- Applications in data science.

8. Decision Trees / CART

- CART algorithm with Gini impurity.
- Constructing decision trees.

9. Logistic Regression

- Derivation and applications in classification.

10. K-Means Clustering

- Step-by-step iterations with dataset.
- Limitations of K-Means.

11. Time Series Analysis

- ARMA/ARIMA models.
- Stationarity and autocorrelation.
- AIC/BIC model selection.

12. Causation vs Correlation

- Difference between correlation and causation.
- Granger causality test.
- Directed Acyclic Graphs (DAG).

Exam Strategy

- If you're short on time, **focus on these 12 topics first** — they cover 80–90% of repeated questions.
 - Then, review the **short notes section** (Empirical Distribution, Statistical Significance, Power-law distribution, ACF/PACF) since they often appear as smaller but frequent questions.
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